Tidey | for Tidewell, a fictional bank

Case study · Banking · AI agent

A fraud alert agent that knows when to ask

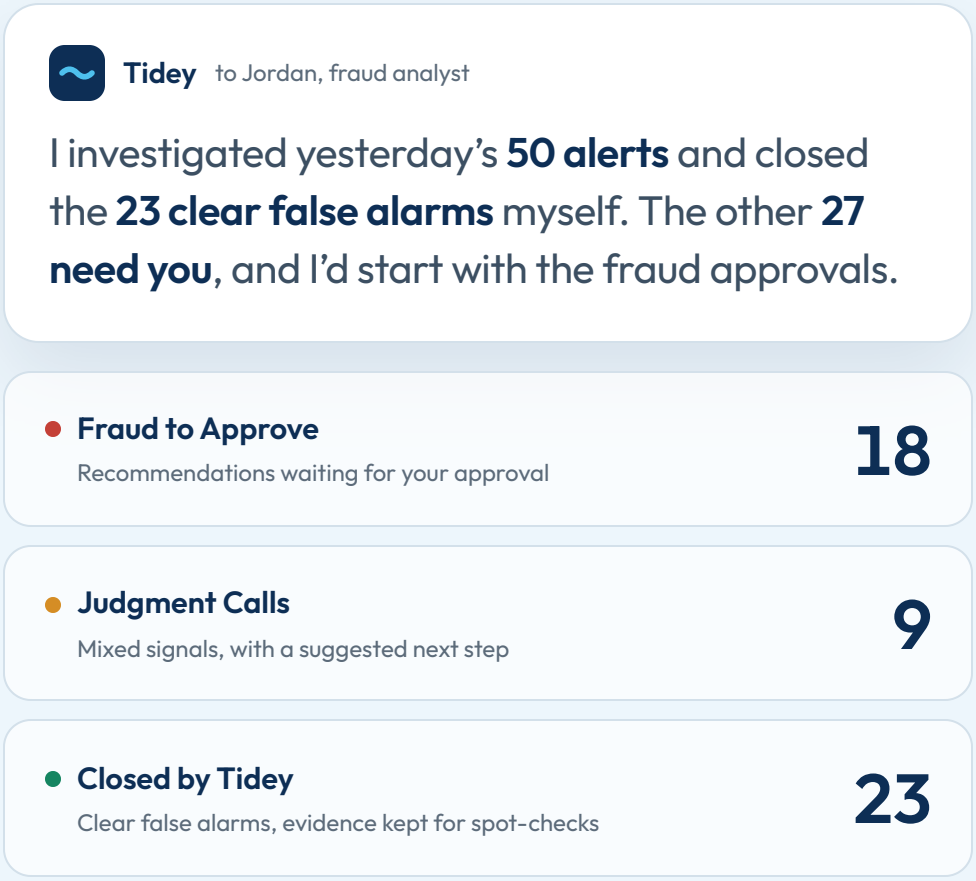
Tidey investigates bank fraud alerts, closes the clear false alarms and hands everything else to an analyst with a recommendation. Built on Jev and tested head to head against a single AI model.

45/50
right on alerts it had never seen

0
fraud alerts closed with nobody looking

3×
cheaper per alert than one AI model

Dan Mota
AI Product Builder · Self-initiated, October 2026



How Tidey opens the analyst's queue after sorting the 50 test alerts. All alerts, customers and transactions are synthetic.

Summary

I designed and built Tidey, an AI agent that investigates fraud alerts for Tidewell, a fictional bank. It gathers the evidence, sorts every alert into **fraud to approve**, **a judgment call** or **a false alarm it closes itself**, and writes the analyst a summary. A person approves every block. To test the design, I built the same agent a second way, with one AI model doing everything, and scored both on 50 alerts neither had seen.

45/50
correct decisions (one AI model: 46)

0
fraud alerts closed with nobody looking (one AI model: 0)

2
unclear alerts closed alone (one AI model: 4)

$0.0012
per alert, summary included (one AI model: $0.0037)

0.30 s
to a decision (one AI model: 2.2 s)

What I built

The agent.
Code, Jev and an AI model each do the part they're best at, and plain code rules decide what happens.

The analyst's app.
A queue that says what's waiting, a case page with the evidence, and one decision bar.

The test.
Two synthetic sets of 50 alerts with an answer key, a scorer, and a live run of five sample alerts anyone can start.

The proof pages.
Every result and every decision is public, down to the probabilities.

What the test showed

About as accurate, with different mistakes.
Tidey sent 3 legitimate alerts to a person it could have closed. The single model closed 4 unclear alerts that a person should have seen. Tidey closed 2.

A third of the cost, 7× faster.
Jev answers short questions instead of one model reasoning through every alert.

Every call traces to a rule.
The lines are numbers in code, so a bank can see them, audit them and move them.

All alerts, customers and transactions are synthetic, generated with an answer key. Tidewell is a fictional bank. Jev is TypeSafe AI's decision model; the single model is OpenAI GPT-5.4 mini.

01

The problem

Every fraud alert needs the same groundwork before anyone can decide: who the customer is, what's normal for them, what changed and when. Analysts do that groundwork by hand, alert after alert.

10–15
minutes, the typical time allowed for a retail fraud alert

6–8
systems an investigator checks in a single review

30–40%
of retail alerts need a call to the customer

From interviews with fraud teams at banks in North America and Europe (NICE Actimize, 2026). NICE Actimize sells fraud software, so these are vendor figures.

Before anyone decides, someone has to piece the story together.

The bet

An agent can do the groundwork and the clear calls, so analysts spend their time on the alerts that need a person. The hard part isn't making an agent decide. It's making it know when not to.

What the agent had to be

Careful
Never closes a risky alert alone
Fraud or mixed signals always reach a person.

Accountable
A person approves every block
Tidey recommends. It never freezes an account by itself.

Explainable
Every call traces to a rule
An analyst can see the evidence and the exact line it crossed.

Cheap
Fractions of a cent per alert
Cheap enough to run on every alert, not just a sample.

Where Tidey fits

An alert fires
A bank rule flags a new payee, a far-away purchase or a new device.

→

Tidey investigates
Runs the checks, answers the questions, applies the rules.

→

Fraud to Approve
A recommendation, waiting for approval

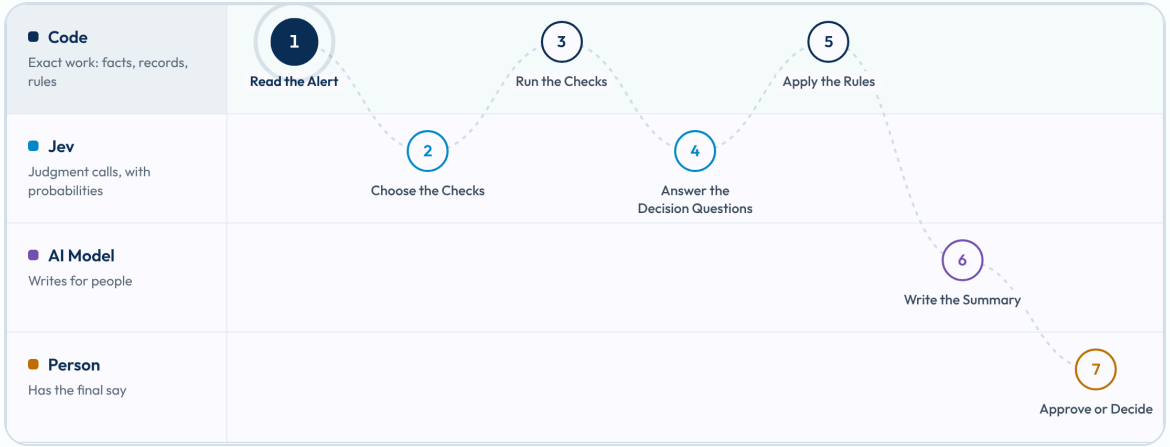
Judgment Calls
Mixed signals, with a suggested next step

Closed by Tidey
Clear false alarms, evidence kept

02

The idea: split the decision

Ask one model to investigate and decide, and you get a single answer with no line you can set or audit. Tidey splits the work by what each part is good at, and keeps the lines in plain code.



Seven steps across four lanes. Code does the exact work, Jev makes the judgment calls with probabilities, an AI model writes for people, and a person has the final say.

Why Jev

Jev answers a yes/no or pick-one question with a probability for each answer. That turns “is this fraud?” into a number a bank can draw a line on, and move without rewriting a prompt.

Why not one model

A single model returns its decision and a confidence it wrote itself. You can’t set where it closes an alert, only ask it nicely.

The lines live in code

Scale: how likely Jev thinks the activity is legitimate

Escalate as Fraud	Send to a Person	Close It Alone
A fraud type ≥ 0.85	Everything that misses a rule, with a summary and a suggested next step	Legitimate ≥ 0.85

0 0.15 0.85 1

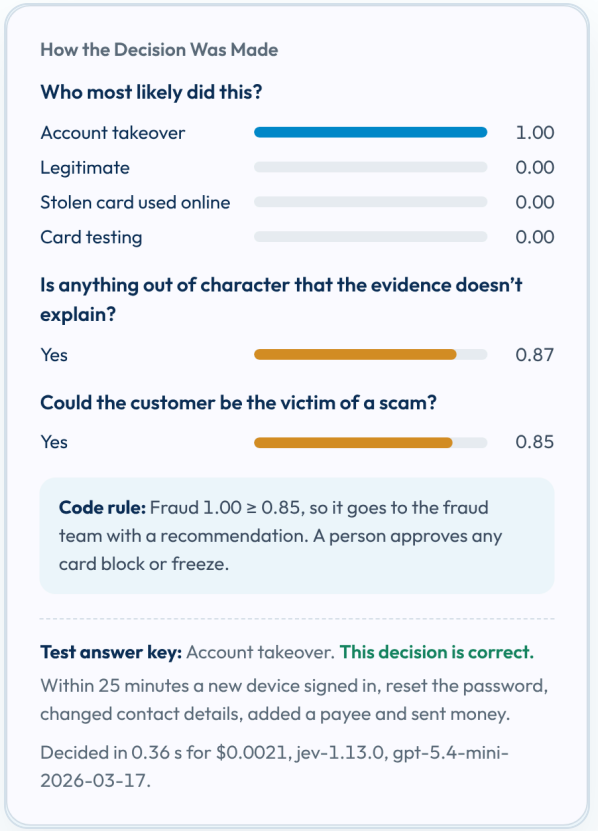
To close alone, Tidey also needs the chance that something is out of character and unexplained at 0.35 or less, a scam risk under 0.6, and an amount under $5,000. Escalations are recommendations: a person approves any block. Jev also decides which optional checks to run, at 0.5.

03

How it works: one alert, end to end

Alert A-1007 from the official test run: a password reset, a new payee and a $3,600 transfer at 2:30 AM.

1	**Read the alert** Code	Works out the facts first: **55× the customer's usual $65.34**, 833 miles from home in San Diego, overnight, from a Windows laptop first seen that day.
2	**Choose the checks** Jev	Transfers always get transaction history, payee check and account changes. Jev adds devices and sign-ins 0.91, runs and skips past alerts 0.49 and merchant check 0.31.
3	**Run the checks** Code	Password reset at 2:12, email changed at 2:16, payee added at 2:24, money sent at 2:30, **6 minutes after adding the payee**. No earlier transfers to them.
4	**Answer the decision questions** Jev	Who most likely did this? **Account takeover 1.00.** Anything out of character and unexplained? **0.87.** Could it be a scam? **0.85.**
5	**Apply the rules** Code	Fraud 1.00 ≥ 0.85, so it goes to the fraud queue with a recommendation.
6	**Write the summary** AI model	"Approve freezing online banking and holding the transfer. Check whether the password reset, email change, new device sign-in, and payee addition were authorized before releasing funds."
7	**Approve or decide** Person	The analyst approves the freeze. Until then, nothing happens to the account.

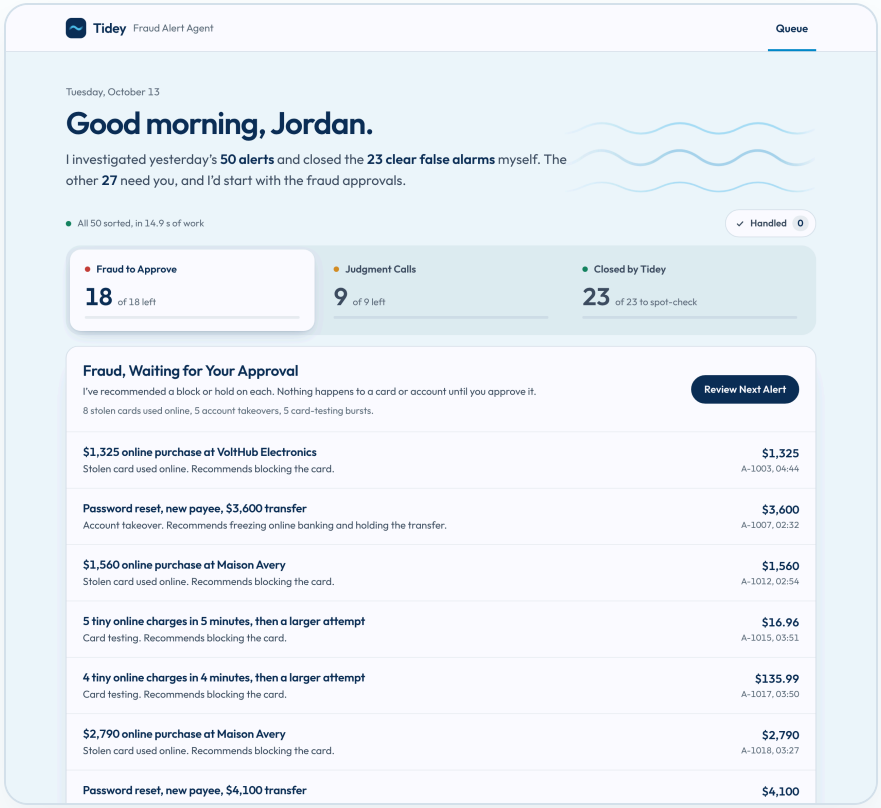


The case page shows the math. Jev's probabilities, the rule that fired, and the answer key for the test. Decided in 0.36 s, for $0.0021 including the summary.

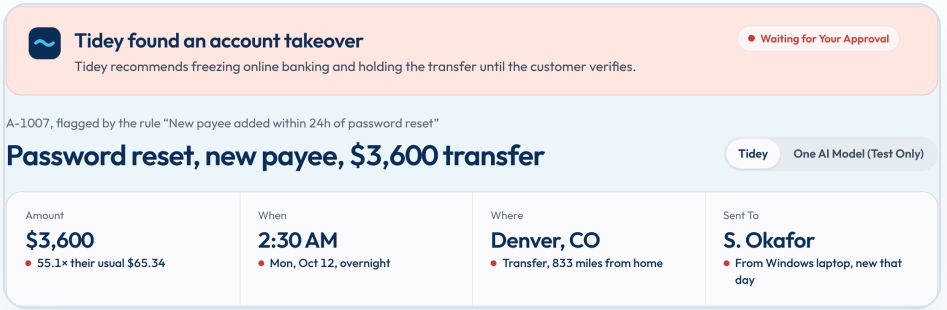
04

The analyst's side

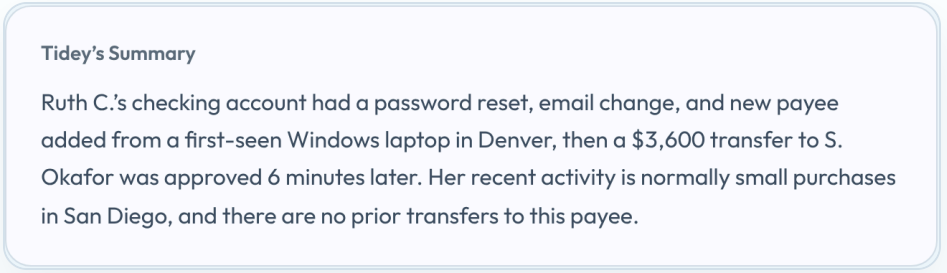
Analysts only ever use Tidey. The queue says what's waiting and where to start, and every case ends in one decision bar.



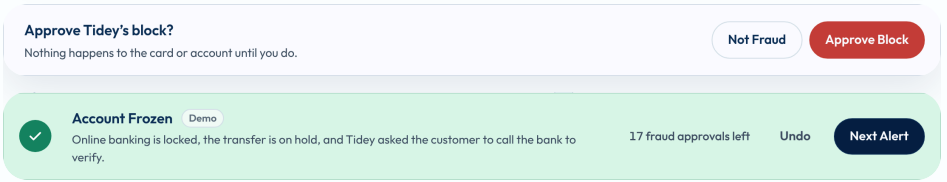
The queue. Tidey says what it did overnight and splits the rest into three groups: Fraud to Approve, Judgment Calls and Closed by Tidey. Closed alerts stay in the queue for spot-checks.



The case page. Tidey's call comes first, then the facts that matter most: amount, time, place and device or payee.



A summary written for the analyst, with the evidence trail and the math below it.



One decision bar. The buttons change with the group: Approve Block or Not Fraud for fraud, Mark Legitimate, Escalate as Fraud or Contact Customer for judgment calls, and Looks Right or Reopen for closed alerts.

05

Design decisions

Six choices shaped how Tidey decides and how far it's trusted.

Questions, not one decision

Probabilities make every line tunable

Jev answers narrow questions. Code combines the answers, so each threshold can be read, tested and moved on its own.

Rules in code, not prompts

Policy stays policy

Changing where Tidey closes an alert is a one-line change a fraud lead can review, not a prompt rewrite that might shift everything else.

Recommend, don't act

A person approves every block

Freezing an account hits a real customer and is hard to undo. Tidey recommends and explains; an analyst decides.

A hard cap

Never close alone at $5,000 or more

However confident Tidey is, large amounts always reach a person. The single model gets the same guardrail.

Checks on demand

Jev picks the optional checks

The checks every alert of that type needs always run. Jev decides whether the rest are worth running, which keeps each investigation short.

Write for people only

A summary only when a person will read it

Closed false alarms keep their evidence but skip the AI-written summary, which is most of the cost.

A fair comparison

The single model gets the same six checks as tools, the same definitions of each fraud type, the same $5,000 guardrail and the same alerts. The only difference is who decides: one model, or Jev plus code rules.

Nothing hidden

Both versions log every step. The test results page shows every alert, both decisions and the answer key, and anyone can rerun the five sample alerts live.

06

The test

Two sets of 50 synthetic alerts, each with an answer key neither version sees. I tuned Tidey on the first set, then ran both versions once on the second.

What's in each set

24	**Legitimate.** False alarms: travel, new devices, big but normal purchases
18	**Fraud.** 8 stolen cards used online, 5 account takeovers, 5 card-testing bursts
8	**Unclear.** Mixed signals a person should look at

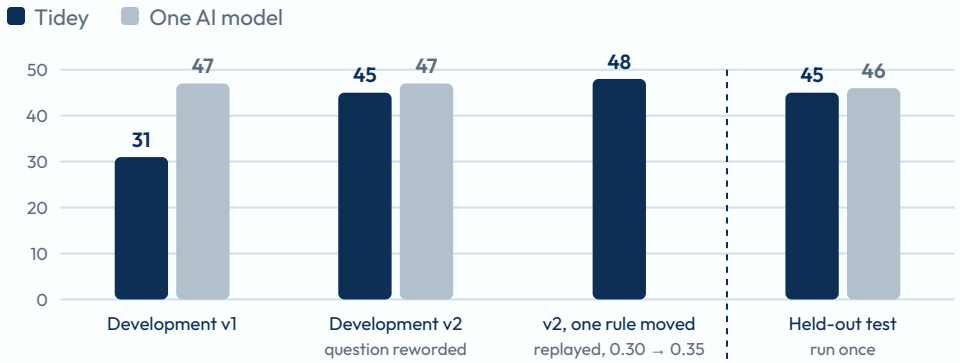
How a decision is scored

Right	Close a legitimate alert, escalate fraud, send an unclear alert to a person
The costly mistake	A fraud or unclear alert closed with nobody looking
The cheap mistake	A legitimate alert sent to a person for an extra review

Why a second set

Tuning on the alerts you report on flatters the results. The second set was generated after tuning, with a new seed, new places, merchants and amounts, and 8 new unclear cases.

Correct decisions out of 50



v1: 31/50.

My first question was the wrong one. I asked "is everything explained?", and almost nothing is fully explained, so 19 legitimate alerts went to a person.

v2: 45/50.

Turned around to "is anything out of character and unexplained?" One question, 14 more right.

Then one rule moved.

Three legitimate alerts sat just over the 0.30 limit; the closest risky one was at 0.46. Moving the limit to 0.35 replayed at 48/50.

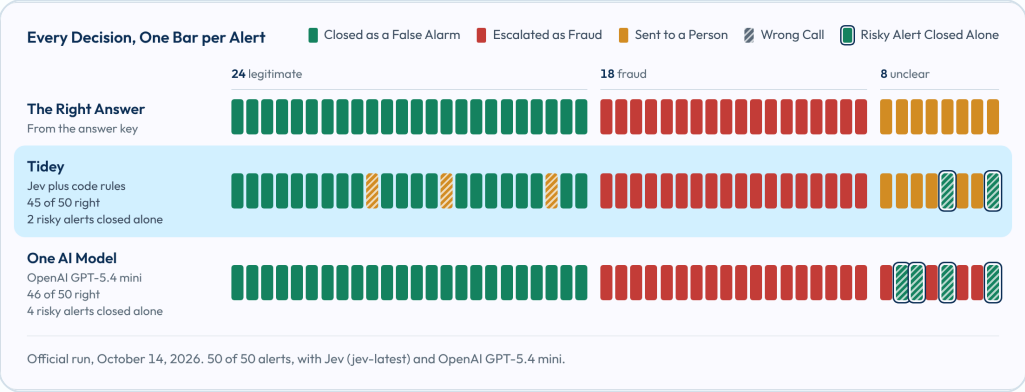
Held-out test: 45/50.

Run once on the second set. Nothing was re-tuned after it.

07

Results on the held-out set

About as accurate overall, with different mistakes. Tidey trades **3 extra reviews** for **half the risky closes**, at **a third of the cost** and **7× the speed**.



Every decision, one bar per alert. Hatched bars are wrong calls; outlined bars are risky alerts closed with nobody looking. Tidey's three amber hatches are legitimate alerts it sent to a person.

> **Which trade is right is a policy choice**
>
> A bank that can't afford a missed skimming case wants Tidey's side. A bank drowning in reviews might move the line. In Tidey that's one number in code, tested before it ships.

Measure	Tidey	One AI model
Correct decisions	**45 / 50**	46 / 50 better
Fraud closed with nobody looking	**0**	0
Unclear alerts closed with nobody looking	**2** better	4
Unclear alerts a person saw	**6 / 8** better	4 / 8
Legitimate alerts sent to a person	**3**	0 better
Right fraud type when escalated	**18 / 18**	18 / 18
Cost per alert	**$0.0012** better	$0.0037
Time to a decision	**0.30 s** better	2.2 s
Broken or invalid outputs	**0**	0

Official run, October 14, 2026: Jev jev-1.13.0 and OpenAI GPT-5.4 mini. Tidey's cost includes the summary; its summary is written after the decision, for alerts a person reads. The single model decides and writes in one call.

08

Where a person stays in the loop, and the two misses

Tidey is built to hand off. These are the points where a person always decides, and the two alerts that got past them.

Point	Who	Why
Every block or freeze	Analyst	It hits a real customer and is hard to undo. Tidey only recommends.
Judgment calls	Analyst	Mixed signals or scam risk. The rules send anything below the lines to a person.
$5,000 and over	Analyst	Never closed alone, however confident Tidey is.
Closed alerts	Analyst, spot-check	Every close keeps its evidence. Reopen sends it for a second look.
The lines themselves	Fraud lead	Thresholds are policy, set and reviewed in code.

The two misses

A-1036

$600 ATM withdrawal in Cancun

A booked trip, but a non-bank ATM at 2:40 AM is a common skimming pattern.

Legitimate	**1.00**
Unexplained	**0.08**

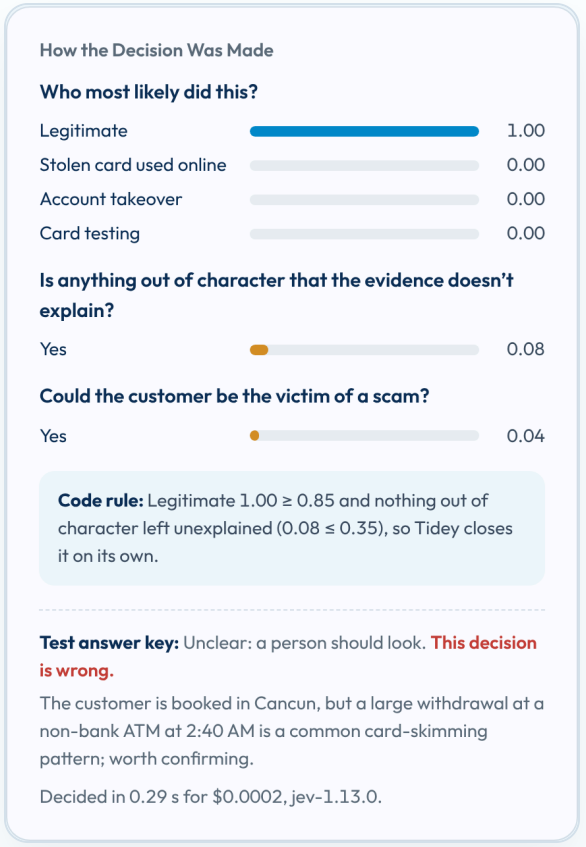
A-1049

$64 at a usual bookstore

From a new phone that took three tries at two-factor while the old one was still in use.

Legitimate	**0.99**
Unexplained	**0.10**

The single model closed both of these too, plus two more unclear alerts.



A-1036 on the case page. Every rule passed. Jev answered the question it was asked, and the answer key says a person should have looked.

09

What the test taught me

Both misses passed every rule. Jev answered the question I asked, who most likely did this, and for both the customer was a fair answer. I never asked whether the pattern is risky even when the customer can explain it. **The miss was the question I didn't ask**, and the fix is a new question, not a stricter number.

What I learned

The question matters more than the threshold.
Asking the opposite question took Tidey from 31 to 45 right.

Keep a held-out set.
48/50 replayed on the tuning set became 45/50 on fresh alerts. Without the second set I would have reported the 48.

Splitting the decision paid twice.
It made Tidey easier to control, and also cheaper and faster.

What I'd do next

A third question for Jev.
"Is this risky even if the customer can explain it?", for patterns like overnight cash at a non-bank ATM or a new device that failed two-factor. A high answer goes to a person.

Re-test on a fresh set.
So the rules are checked on alerts they weren't written for.

Real labeled alerts, then shadow mode.
Calibrate the lines on real data, run next to analysts, and measure the minutes saved per alert.

What this project shows

Agent design
Code, a decision model and an AI model, each doing what it's best at, with the lines a bank cares about in code.

Honest testing
A head-to-head comparison, a held-out set and every result published, including the misses.

Product judgment
An agent that knows when to ask, and an analyst app built around the moment a person decides.

Try it: tidey-fraud-agent.vercel.app · **Code:** github.com/dandmxstudio-droid/tidey-fraud-agent

Sources: NICE Actimize, "Beyond the Alert Screen: Research-Based Insights on Rebuilding Fraud Investigations for AI" (white paper, July 2026). TypeSafe AI, Jev documentation (docs.typesafe.ai). Test results: Tidey official run, October 14, 2026 (public/results/official.json in the repo). All alerts and customers are synthetic; Tidewell is fictional.